



Laxmi Charitable Trust's
Sheth L.U.J. College of Arts & Sir M.V. College of Science & Commerce
Dr. S. Radhakrisnan Marg, Andheri (East) Mumbai 400 069



Department of Information Technology (Bsc.IT Semester III)
Scala for DS

Practical - 28

Roll No. : 44	Name: Amit Umarvaish
Class : SYIT	Date/Time of Submission : 22-09-2026

```
// 28) Compute the Euclidean distance between two Breeze  
vectors. Use it for  
// nearest neighbor classification.
```

Code:

```
import breeze.linalg._  
  
object Prac27 extends App  
{  
  val A = DenseVector(2.0, 3.0)  
  val B = DenseVector(5.0, 7.0)  
  
  val distance = norm(A - B)  
  
  println("Euclidean Distance: " + distance)  
  
  val C = DenseVector(2.5, 3.5)  
  
  val d1 = norm(C - A)  
  val d2 = norm(C - B)  
  
  if (d1 < d2)  
    println("Nearest Neighbor: A")  
  else {  
    println("Nearest Neighbor: B")  
  }  
  println("Amit 44")  
}
```

Output:

```
"C:\Program Files\Java\jdk1.8.0_181\bin\java.exe" ...
```

```
Euclidean Distance: 5.0
```

```
Nearest Neighbor: A
```

```
Amit 44
```

```
Process finished with exit code 0
```